AI-BASED ELECTRICITY CONSUMPTION PREDICTION & ALERT SYSTEM

1M1B Green Skills And Applied AI Internship

Presented by: Shruti Das

PROBLEM STATEMENT

Problems faced:
- High electricity consumption
- Electricity wastage
- Power shortages
- High electricity bills
- Uneven power distribution

Affected:
- Residential consumers
- Educational institutions and students
- Industrial and commercial sectors

Importance of the solution:
- Improved power distribution.
- Reduced electricity wastage.
- Early detection and alerting regions with sudden spike.

Real world examples:
- During hot weather, increased AC usage can lead to much higher electricity consumption.
- Many a times, the lights and fans are left ON even if they are unoccupied.

RESEARCH & INSIGHTS

- Collected a 21-day household electricity consumption dataset.
- **Data collected:** Temperature(°C), Season/Weather, AC hours, Fan hours, Light hours, Daily electricity consumption(kWh).
- Higher AC usage considerably increases the daily consumption.
- Temperature influences the AC or fan usage just like the duration of the day influences light hours.
- Daily electricity consumption can be predicted using appliance usage patterns and temperature. This insight led to the development of a Machine Learning model that predicts household electricity consumption.

AI-ENABLED SOLUTION

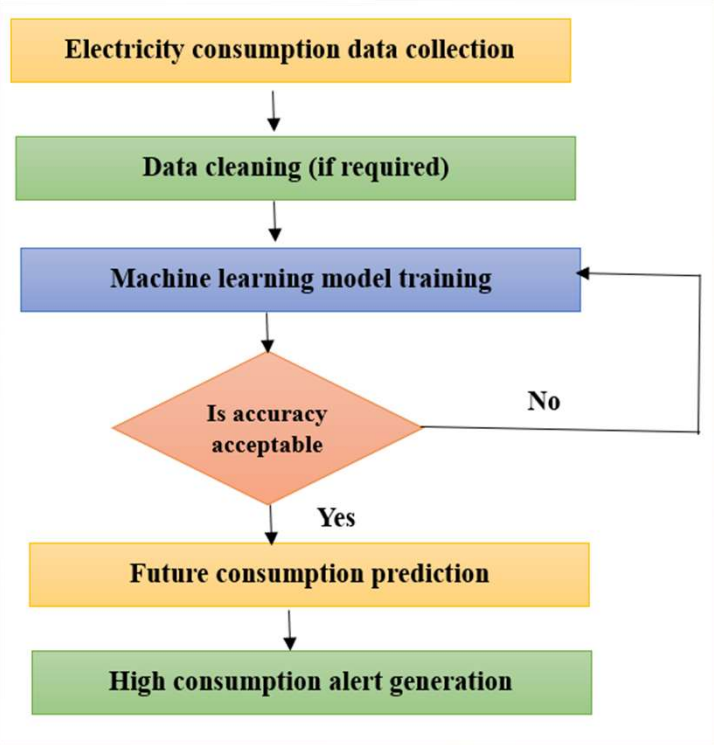
- **Solution:** The system collects and cleans household electricity consumption data, then trains a Machine Learning model to predict future electricity usage. If the model achieves acceptable accuracy, it predicts future consumption and generates alerts for sudden spikes in electricity usage.

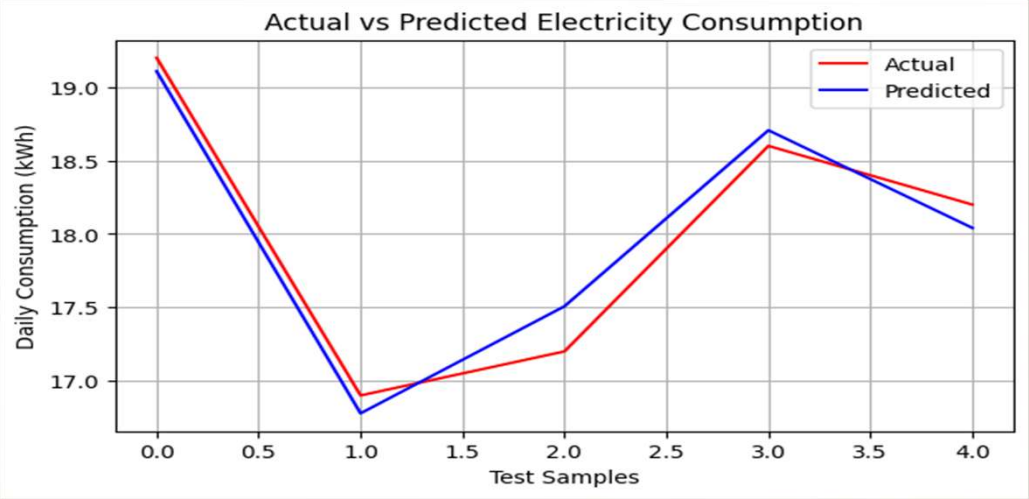
- A **Linear Regression** model will learn the relationship between the features and target variable and ultimately predicts the expected daily consumption.

- **Tools Used:** Python, Pandas, Google Colab, Scikit-learn, Matplotlib, Microsoft Excel

- **Key Features:**
  - ✓ Predicts daily electricity consumption using appliance usage and temperature as inputs.
  - ✓ Displays actual vs predicted results.
  - ✓ Generates alerts and messages for various consumption value range.

PROTOTYPE TESTING & RESULTS





Date	Season	Temperature(°C)	AC hours	Fan hours	Light hours	Daily Consumption(kWh)
04-07-2026	Rainy	33	8	18	7	19.2
05-07-2026	Rainy	32	8	16	7	18.6
06-07-2026	Rainy	30	7	17	6	16.8
07-07-2026	Rainy	31	6	18	7	17.5
08-07-2026	Rainy	33	7	16	6	17.9
09-07-2026	Rainy	31	5	19	8	16.4
10-07-2026	Rainy	29	5	20	5	15.7

Code & CSV file: https://drive.google.com/drive/folders/1m7mQV8jKGloaocisbfScBVP2A1hBOicr

OUTCOME & IMPACT

Solution:

- The solution was tested in **Google Colab** using a 21-day household electricity consumption dataset.
- The **Linear Regression** model was trained and tested using a train-test split of 80:20.
- The predictions were compared with the actual values.

Key Results:

- The model predicted the daily electricity consumption.

```
Predicted Daily Consumption: 20.17 (kWh)
```

- An R^2 score of 0.96 and Mean absolute error of 0.16 was derived.

```
# Evaluation
print("R2 Score:", round(r2
print("Mean Absolute Error:

R2 Score: 0.96
Mean Absolute Error: 0.16
```

Challenges:

- Monitoring and noting down daily electricity consumption data.
- Ensuring the data was correctly formatted before training the model.

Feedback/Improvements:

- Increasing the dataset size.
- Testing the dataset on more ML models

FUTURE SCOPE

- Expanding the dataset size and also improving the prediction accuracy by testing additional machine learning algorithms.

- The solution can be scaled from a single household to multiple households, localities, and cities.